UNIT - V

Handwritten Study Notes | JNTU(A) R23

Syllabus Topics:

Rain water management | Planning and operation of irrigation systems | Conjunctive use of water | Participatory irrigation management and WRM | Water management policy during droughts | Predicting effect of water shortage on crops | Water footprint of crops and its applications | Blue, green and grey water footprint

1. Rain Water Management

Rain water management: planned handling of rain (the primary source of water) so that maximum useful water is stored in the soil, ponds and aquifers and the excess is drained safely. About 55 % of India's net sown area is rainfed, so it is the Key to food security.

Components

* In situ conservation: store rain in the soil profile - contour farming, bunds, broad bed and furrow, mulching, tillage, tied ridges, land levelling, deep ploughing.
* Ex situ harvesting: runoff stored in farm ponds, check dams, percolation tanks and used for supplemental irrigation.
* Recharge: rooftop and recharge pits, trenches, injection wells; Managed Aquifer Recharge (MAR).
* Safe disposal: grassed waterways, diversion drains, drop structures, check dams to carry excess flow without erosion.
* Crop management: suitable crops and varieties, sowing at onset of monsoon, mid-season correction, mulching, intercropping, weed control, anti-transpirants.

Effective rainfall = Rainfall - Runoff - Deep percolation - Interception loss

* Effective rainfall methods: USDA-SCS (monthly), FAO/AGLW formula: Pe = 0.6 P - 10 (P < 70 mm/month); Pe = 0.8 P - 24 (P > 70 mm/month).
* Dependable rainfall: 75 % probability rainfall (rain expected 3 of 4 years) used in planning (Weibull plotting position m / (n+1)).
* Water Use Efficiency WUE = yield (Kg/ha) / water used (mm or m3). Rain-water use efficiency RWUE = yield / rainfall.

Remember:

Rain water management = capture + store + use efficiently + recharge.

Practice: Conserve in the soil first, then store extra runoff in ponds for protective irrigation.

2. Planning and Operation of Irrigation Systems

Irrigation: artificial application of water to soil to supply moisture essential for plant growth. Planning gives the right project, operation gives right use.

Planning of an irrigation system

1. Water availability: surface (river, reservoir, tank), groundwater, 75 % dependable yield, losses, other uses; reservoir simulation.
2. Command area and soils: Gross Command Area (GCA), Culturable Command Area (CCA) = GCA - unculturable; Net sown area; land capability and drainage.
3. Crop water requirements: crop pattern and calendar, ETO and Kc, effective rainfall, losses, leaching.
4. Selection of system: gravity canal, lift, tank, well, drip, sprinkler; storage + diversion headworks.
5. Design of canal network: main canal, branches, distributaries, minors, water courses and field channels; capacity from peak demand (duty).
6. Economic analysis: benefit-cost ratio, IRR, financial viability, social and environmental impact.

Key terms and formulas

$$ETc = Kc \times ETO \; ; \quad NIR = ETc - Pe \; ; \quad GIR = NIR / Ea$$

(Ea = field application efficiency)

* Duty (D): area irrigated per unit discharge (ha per cumec), Delta (Delta): total depth of water required by crop in base period B (days).

$$Delta = 8.64 \; B / D$$

(Delta in m, B in days, D in ha/cumec)

* Base period (B), Kor depth/period, Kor watering for first irrigation, Consumptive use Cu, Irrigation interval = (Field capacity - Allowable depletion) / ETc.
* Efficiencies: conveyance Ec = water delivered to field / water at head; application Ea = water stored in root zone / water delivered; project efficiency = Ec x Ea (typical canal 30-40 %, drip 85-90 %).

Method	Efficiency	Suited for
Surface (border, furrow, basin)	40 - 60 %	level fields, cereals, rice
Sprinkler	70 - 80 %	undulating land, pulses, vegetables
Drip / micro	85 - 95 %	orchards, row crops, water scarce area
Sub-surface / pitcher	90 %	arid, small gardens

3. Operation of Irrigation Systems

* Water delivery methods: continuous flow, rotational (Warabandi - North India, fixed turn by time in proportion to land holding), demand supply, arranged delivery.
* Warabandi: fixed weekly roster - each farmer gets water turn (hours = area x time per ha), ensures equity at tail end.
* Canal operation: flow regulation at head regulators and cross regulators, scheduling of canal closures, monitoring discharge with weirs/flumes,

SCADA/automation, volumetric supply.

* Reservoir operation: rule curve - release as per demand and flood storage, conjunctive operation with wells.
* Maintenance: de-silting, repair of lining, weed removal, dividing structures, drainage clearing, Operation and Maintenance (O and M) funded by water charges.
* Problems: tail-end deprivation, seepage and waterlogging, salinisation, low water-use efficiency, unauthorised drawing, poor O and M funds.
* Modernisation: lining, pressurised pipe network, Pressure irrigation, on-line monitoring, remote sensing for crop area and ET, SCADA, farmers' water-user associations.
* Irrigation scheduling: when and how much - based on soil moisture (tensiometer, gypsum block), climatic approach (IW/CPE ratio = 0.8 to 1.0), plant indicators, critical growth stages (crown root, flowering, grain filling).

> Tip:
> Warabandi and rotational supply deliver water in proportion to land - transparent and equitable, combine with volumetric charges to save water.

4. Conjunctive Use of Water

Conjunctive use: the coordinated and planned use of surface water and groundwater so that the total yield is greater than separate use, with minimum environmental damage.

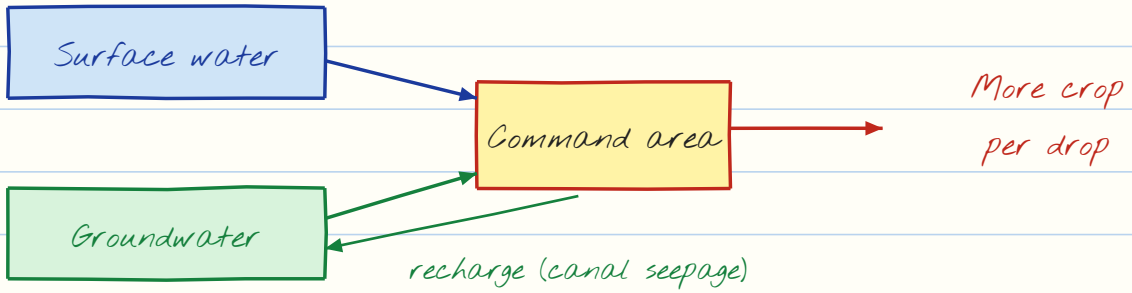


Fig 1: Conjunctive use idea

Why?

* Surface supply is seasonal, groundwater is available throughout the year but depends on recharge. Using both reduces risk.
* Canal irrigation causes waterlogging in head reach and shortage in tail reach; pumping lowers the water table (bio-drainage) and relieves waterlogging.
* Wells can supplement canal supply at tail and in droughts; canal water recharges the aquifer through seepage.

Methods

* Use surface water when plentiful (monsoon) and store excess in aquifer (artificial recharge); pump groundwater in lean season and drought.
* Blend saline groundwater with fresh canal water (cyclic or blending) for irrigation.
* Tubewell in canal command area (canal + well), tank + well (south India), conjunctive operation of surface reservoir and aquifer storage (Aquifer Storage and Recovery - ASR).
* Simulation and optimisation models (linear programming, MODFLOW + reservoir models) determine optimal pumping and surface supply.

Advantages and concerns

* Advantages: higher cropping intensity, more reliable supply, control of waterlogging and salinity, lower storage evaporation, saves canal capacity, cost-efficient.

* Concerns: over-draft if recharge is not planned, quality of groundwater (fluoride, arsenic, salinity), energy for pumping (subsidised power), institutional problem (two departments - one for canal and one for groundwater).

5. Participatory Irrigation Management (PIM)

PIM: involvement of water users in all aspects of irrigation management - planning, design, construction, operation, maintenance, monitoring and financing - by transferring responsibility to farmer organisations.

* Water Users Associations (WUA) at the water-course/minor level; Distributory Committee and Project Committee above it.
* Laws: Andhra Pradesh Farmers' Management of Irrigation Systems Act (1997) - first major reform, Maharashtra Management of Irrigation Systems by Farmers Act 2005; Pani Panchayats (Maharashtra, Odisha), Water Users Societies; Tamil Nadu WUA Act 2000; Joint Irrigation Management (JIM) - Gujarat Sinchai Mandali.
* Functions: distribute water equitably, collect water charges, maintain channels, resolve conflicts, plan cropping pattern, record keeping and share information.
* Benefits: efficient and equitable distribution, higher recovery of water charges, better maintenance, reduced conflicts, tail-end water reaching, improved yields and ownership.
* Constraints: lack of legal and financial powers, political interference, caste/land inequalities, weak training and technical support, poor condition of the physical system at hand-over.

Success:

PIM works when: users have legal rights and clear water entitlement, canal is in good condition, farmers get capacity building, and the government supports with funds.

6. Integrated Water Resources Management (IWRM)

IWRM (Global Water Partnership, 2000): a process which promotes the coordinated development and management of water, land and related resources in order to maximise economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems.

Principles

* Dublin principles (1992): (1) fresh water is finite and vulnerable resource; (2) development and management should be participatory; (3) women play a central role in provision, management and safeguarding; (4) water has an economic value and should be recognised as an economic good.
* Three E's: Efficiency (use water wisely), Equity (fair access for all), Environmental sustainability (leave water for ecosystems).

Elements (framework)

Pillar	Contents
Enabling environment	policies, laws, financing, national water plan
Institutional roles	river basin organisation, local bodies, WUAs, NGOs, private sector
Management instruments	water resource assessment, demand management, allocation, pricing, information, conflict resolution, regulation, monitoring

* River basin or watershed as the planning unit, link surface and groundwater, quantity and quality, upstream and downstream, land and water, green and blue water.
* Cross-sector integration (agriculture, industry, domestic, energy, environment) and integration across scales (local to national).
* In India: National Water Policy 2012, River Basin Organisations (Brahmaputra Board, Ganga basin authority), Water Resources Information System (India-WRIS), Jal Jeevan Mission, National Water Mission under NAPCC.

7. Water Management Policy During Droughts

Drought types: meteorological (rain deficit), hydrological (low streamflow/groundwater), agricultural (soil moisture deficit), socio-economic. IMD: drought if seasonal rain < 75 % of normal (rain deficiency > 25 %), severe if > 50 % deficiency.

Policy components

* Preparedness: drought monitoring and early warning (rain, SPI, NDVI, soil moisture, reservoir storage), vulnerability maps, contingency crop plans (CRIDA), drought manual 2016 (revised 2020).
* Water priority during drought: 1) drinking and domestic, 2) livestock, 3) protective irrigation for standing crops/perennial, 4) industry, 5) other. Cut allotment to low-value uses, ban water-intensive crops, restrict pumping and tanker supply in regions.
* Supply side: use dead storage with care, deepen wells and tanks, tanker supply, emergency recharge, water transport, desalination, treated waste water reuse.
* Demand side: rotational supply, volumetric limits, mulching, drip, water pricing, crop shift to pulses/millets, fallowing.
* Relief and livelihoods: MGNREGA work for water conservation, fodder banks and cattle camps, drinking water, crop and livestock insurance (PMFBY), seed supply, credit relief, public distribution.
* Long-term: drought-proofing - watershed development, tank desilting, groundwater recharge, farm ponds, afforestation, diversification, inter-basin transfer where required.

> Concept:
> Drought management cycle = prepare -> monitor -> respond -> recover -> mitigate. Shift from 'crisis management' (relief) to 'risk management' (preparedness).

8. Predicting Effect of Water Shortage on Crops

Crop yield falls when ET is less than potential (crop evapotranspiration ETc). The shortage effect depends on the growth stage and how severe the stress is.

$$(1 - Ya/Ym) = Ky\ (1 - ETa/ETm)$$

(Doorenbos & Kassam, FAO-33)

Ya = actual yield, Ym = maximum yield, ETa = actual ET, ETm = max ET, Ky = yield response factor (crop and growth stage specific).

Crop	Ky (whole season)	Critical stage
Maize	1.25	flowering / silking
Wheat	1.0 - 1.15	crown root, flowering
Rice	1.1 - 1.3	flowering, panicle initiation
Cotton	0.85	flowering, boll formation
Groundnut	0.70	flowering, pegging
Sugarcane	1.2	grand growth

* Ky > 1: sensitive (yield drops more than proportionally); Ky < 1: tolerant crop. Total shortage multi-stage product model: Ya/Ym = product of (1 - Kyi (1 - ETai/ETmi)) over stages.

Example

Maize, ETm = 500 mm, ETa = 400 mm, Ky = 1.25: 1 - Ya/Ym = 1.25 x (1 - 0.8) = 0.25 -> Ya = 75 % of Ym. A 20 % ET deficit gives a 25 % yield loss.

* Other approaches: crop growth simulation (AquaCrop of FAO, DSSAT, APSIM, InfoCrop) use soil water balance, canopy and harvest index to predict yield under deficit irrigation.
* Water stress indices: Crop Water Stress Index (CWSI) using canopy temperature, soil moisture depletion fraction p, relative water content, NDVI and remote sensing.

* Consequences: reduced germination, shorter growth, poor flower/grain fill, low biomass, quality loss; Strategy: deficit irrigation (RDI), supplemental irrigation at the critical stage, drought-tolerant varieties, mulching and conservation.

9. Water Footprint of Crops and Applications

Water Footprint (WF) - Hoekstra (2002): the total volume of fresh water used, directly and indirectly, to produce a product, measured over the full supply chain. For crops: m3 per tonne (or litre per Kg).

WF = Blue + Green + Grey (m3 per tonne of crop)

BLUE WF	GREEN WF	GREY WF
irrigation water (surface + ground)	rain water used by the crop (soil)	water to dilute pollutants

Water Footprint of a crop

Fig 2: Components of water footprint

$$WF_{crop} = (CWU_{blue} + CWU_{green}) / Y + WF_{grey}$$

Blue, green and grey water footprint

* Blue WF: surface and groundwater consumed (evaporated, incorporated in product, or not returned to the same catchment). For crops = irrigation ET (blue CWU) / yield.
* Green WF: rain water stored in the soil and used by the crop (rain-fed ET). Important for rainfed agriculture and forest.
* Grey WF: volume of fresh water required to dilute pollutants (fertiliser, pesticide) to meet ambient quality standards.

$$WF_{grey} = (\alpha \times AR) / (C_{max} - C_{nat}) / Y$$

(AR = chemical applied Kg/ha)

alpha = leaching/runoff fraction (e.g. 10 % for N), C_{max} = max acceptable concentration (e.g. 10 mg/L N), C_{nat} = natural conc.

* CWU: crop water use (m3/ha) = 10 x sum of ETc over the growing period (mm), Y = yield (t/ha). WF (m3/t) = CWU/Y.

Product (global avg)	Water footprint
Wheat	about 1800 m3 / tonne
Rice (paddy)	about 2500 m3 / tonne (India higher)
Sugarcane	about 200 m3 / tonne cane
Cotton lint	about 10,000 m3 / tonne
Beef	about 15,400 m3 / tonne

Applications

* Compare crops and regions; select crops with a low WF per Kg of food or per rupee and per calorie.
* Identify hotspots of water stress, evaluate sustainability (is blue WF within local environmental flow?), basin-level water accounting.
* Water-saving measures: increase yield (WF reduces), use drip, mulching, reduce fertiliser, treat waste water, improve irrigation efficiency.
* Virtual water trade: import of a water-intensive product saves local water - used in national water policy (e.g. rice, cotton exports from India).
* Product labelling, corporate water stewardship, national water footprint of consumption, policy for water-wise diets.
* Water Footprint Assessment 4 phases: set goals and scope, accounting, sustainability assessment, response formulation (Water Footprint Network manual).

Memory trick:

Blue = irrigation water, Green = rain stored in soil, Grey = pollution dilution water. WF reduces if yield goes up and losses go down.

10. Solved Examples and Revision

Example 1: Duty and delta

A crop with base period B = 120 days has duty D = 1200 ha/cumec on field. Find delta.

Delta = 8.64 B / D = 8.64 x 120 / 1200 = 0.864 m = 86.4 cm

Example 2: Irrigation requirement

ETc = 5.5 mm/day, effective rain = 0.5 mm/day, Ea = 70 %, Ec = 80 %, area 10 ha.

* Net irrigation = 5.5 - 0.5 = 5.0 mm/day
* Gross at field = 5.0 / 0.70 = 7.14 mm/day; at canal head = 7.14 / 0.8 = 8.9 mm/day
* Volume = 8.9 mm x 10 ha = 0.0089 m x 100,000 m2 = 890 m3/day

Example 3: Water footprint of rice

Green CWU = 4000 m3/ha, blue CWU = 8000 m3/ha, yield = 4 t/ha, N applied 100 Kg/ha, alpha = 0.1, Cmax = 10 mg/L, Cnat = 0.

* Green WF = 4000/4 = 1000 m3/t; Blue WF = 8000/4 = 2000 m3/t
* Grey WF = (0.1 x 100 Kg/ha) / (0.010 Kg/m3) / 4 t/ha = 1000/4 = 250 m3/t

Total WF = 1000 + 2000 + 250 = 3250 m3 per tonne

Quick revision list

* Rain water management: in situ + ex situ + recharge + safe disposal.
* Irrigation: GCA, CCA, duty, delta, efficiencies, warabandi, scheduling.
* Conjunctive use = surface + ground, planned together, avoid over-draft.
* PIM: WUAs with legal and financial powers; NRM: Dublin principles, efficiency-equity-sustainability.
* Drought: monitor, priority to drinking water, contingency plan, MGNREGA, insurance.
* Crop-water shortage: (1 - Ya/Ym) = Ky (1 - ETa/ETm).
* Water footprint: blue + green + grey.

Writing tip:

Exam approach: define -> draw a neat diagram -> write the formula with symbols -> give an example -> mention Indian case or policy.